

GLOBAL

; up · down · / left/right · **ENTER** select · **/DEL** back · Home: 2-col grid, letter jumps, **/I** hop columns · **{ }** page · **b** screenshot · **h** help · **Fn+/Fn+DEL** STOP all radio/rotator control (Help links: **g** glossary+math · **m** user guide · **s** sat history · **t** tech help · **I** learn theory · **f** band plan)

HOME

ENTER opens item; 2-column grid: Satellites, Next Passes, Passes, Track, World Map, Overhead view, Sun/Moon, Space Wx, Activations, AMSAT status, Weather, Grid dist/bearing, Nearby & DX, Location, Update, Settings, Log, Messages, About, Charge/Sleep

SATELLITES

f favorite · **v** favs-only · **/** search ALL of Celestrak → add as auto-updating fav (10 s/2 h courtesy limits) · **n** new GP sat · **x** del added sat · **e** EQX table · **k** OSCARLOCATOR · **3** 3D globe · **2** sat-to-sat · **o** orbital · **y** sim · **t** transponders · **d** 10-day · **i** illum · **s** AMSAT status · **L** share GP over LoRa · **ENTER** passes · right edge: dot = AMSAT heard, square = telemetry, ring = not heard

SAT-TO-SAT (Sats → 2)

Windows when selected sat + a 2nd fav are BOTH above your horizon (start, dur, peak el each). **n** next sat · **r** recompute · **`** back

LIVE GPS POS (Location → v)

DMS lat/lon (max precision) + decimal, alt, grid, speed, course, Rovers/portable. **`** back

3D GLOBE (Sats → 3)

Wireframe Earth, auto-follows selected sat. Graticule, coastline, day/night terminator, QTH, favs, selected sat + footprint + ground-track trail. **g** DX 2nd location (grid) + footprint · **G** clear · **arrows** turn · **ENTER** re-follow · **`** back

EQX TABLE (Sats → e)

Equator-crossing UTC + longitude (W-positive) for OSCARLOCATOR, next 3 days. **/I** scroll · **d** asc/desc node · **r** recompute · **`** back

OSCARLOCATOR (Sats → k)

Live azimuthal-equidistant plot: sub-point, footprint, QTH range ring + full ground track (AOS/LOS). **m** toggle polar (default, auto N/S, flips at equator) / QTH-centered · **`** back

DX DOPPLER TABLE (Mutual → ENTER → d)

RX/TX dial freqs for BOTH stations every 30s across a mutual window. Two lines/step: me (green) + DX (cyan). From the Mutual list, **ENTER** opens a polar detail (me+DX arcs, AOS/LOS/el) then **d** the table (or **d** straight from the list), **t** cycle transponder · **m** mode: true rule / fixed DL / fixed UL · **a** anchor dial (me/DX RX/TX) · **/I** in fixed mode step anchored dial to round 1 kHz (else passband 1 kHz) · header shows pb +/- from center · **/I** scroll · **`** back

NEXT PASSES (favs)

ENTER track · **m** world map · **t** sky-at-a-glance timeline (bars by elev) · **p** rove planner · **w** workable horizon · **s** target search · **r** refresh · **z** deep-sleep until AOS

WORKABLE HORIZON (Next Passes → w)

10-day union of everything workable across all favorites, **s** states list · **d** DXCC list · **g** re-run incl. grids (off by default) · **w** save to /CardSat/workable/ · **`** back

TARGET SEARCH (Next Passes → s)

Pick one state / DXCC / grid; lists every pass over 10 days where it's workable, time-ordered across all favs (sat / date / window / maxEl). **ENTER** polar of that pass · **w** save to /CardSat/search/ · **`** back

ROVE PLANNER (Next Passes → p)

Survey all favorites from a planned grid+time. Form: grid / date / time / +/- hrs / GO. Rows by AOS: sat / AOS / maxEl / states / DXCC. **ENTER** detail (arc + **s/d/g** state/DXCC/grid lists) · **w** save txt · **I** saved plans (**ENTER** view, **/I** scroll, **d** del) · **g** form

PASSES (sel)

* = optically visible. **/I** select · **d** detail · **t/ENTER** track · **n** add TX · **r** recompute · **x** mutual-DX · **v** 10-day chart · **V** visible-pass list · **i** illum · **g** grids · **w** US states · **e** DXCC · **p** print

TRACK (sel)

` exits to previous screen & KEEPS radio/rotator tracking (green RAD/ROT/R+R in header); **r/o** to stop. **m** TUNE/CAL · **d** tune mode (FULL/DLU/LFULL/hold; FULLu=OTR on uplink knob) · **t** next TX · **n** jump to beacon · **c** CTCSS · **N** sat note · **k** CW both legs (linear) · **r** radio · **o** rotator · **p** polar · **a** point-here arrow · **z** big readout · **y** tilt on/off (ADV) · **f** Manual · **I** log QSO · **v** voice memo (SD) · **g** grids · **w** states · **e** DXCC now · **ix2** report Heard (AMSAT) · after LOS **q** (60s) deep-sleep to next pass · **ENTER** save cal · **Fn+u** fully release USB (host stays resident after disengage) is when the serial console returns)

BIG READOUT (z from Track)

Big RX/TX + az/el + tune mode (follows Track). Radio+rotator keep tracking · **/I** tune · **s/x** step/ctr · **m/d** mode · **t** TX · **n** beacon · **k** CW (linear) · **r** radio · **o** rot · **y** tilt · **I** log · **z/`** back

MANUAL (no radio)

Fix one leg, read Doppler freq to tune the other by hand. **u** toggle HOLD/TUNE leg · **/I** passband (linear) · **m** CAL · **t** next TX · **z** big view · **I** log · **p** polar · **g** grids · **w** states · **e** DXCC · **/f** Track

MANUAL BIG (z from Manual)

HOLD/TUNE legs in big digits · **u** swap leg · **/I** tune · **s/x** · **m** CAL · **t** TX · **z/`** back

TRACK · TUNE

/I tune **-/+** · **s** step 100/1k/5k · **x** recenter · tilt to tune if enabled (ADV)

TRACK · CAL

/I downlink · **/I** uplink · **s** step 10/100/1k · **x** zero

WORKABLE GRIDS / STATES / DXCC

Footprint coverage (per-pass union or live now), count on a cyan line: **g** 4-char grids · **w** US states+DC · **e** DXCC (all 340, hybrid polygons+points). On grids, **f** prefix filter (EM/EM2/EM21, upper-cased), **c** clear. **/I** & **{ }** scroll · **`** back

POLAR / PASS DETAIL

Pass detail: **p** polar of this pass. Polar: **I** log QSO · **v** voice memo (SD) · **p/ENTER/`** back

SUN / MOON

Graphic sky-dome (Sun/Moon glyphs by az/el) · **g** graphic/list · **/I** pick Sun/Moon · **o** rotor track on/off · **s** sky sources · **t** transits · **e** EME · **x** stop · **`** back

SKY SOURCES (Sun/Moon → s)

Planets (cyan dots) + strong radio sources (orange +): Cas A, Cyg A, galactic center, Crab, Virgo A, on a sky dome. Antenna-pointing / RF reference. **/I** select · **`** back

EME / MOONBOUNCE (Sun/Moon → e)

Self-echo Doppler per band (50/144/432/1296/10368, topocentric) · range + rate · path degradation vs perigee · galactic sky-noise flag · SUN flag <10° · **m** mutual window · **p** 30-day plan · **o** rotor track Moon · **`** back

GRID DIST/BEARING (menu)

Enter Maidenhead grid → great-circle distance + beam heading (short/long path, km/mi). **g** grid · **q** QRZ → grid lookup (seeds calc) · **o** point rotor at bearing · **`** back

SPACE WX (menu)

Solar 10.7cm flux + planetary Kp + A index + aurora likelihood (from Kp), labeled & color-coded, with HF/sat operating outlook & data age - **p** HF/6m propagation - **m** MUF to world regions - **r** refresh (WiFi) - **⌵** back

HF/6m PROPAGATION (Space WX → p)

Turns solar flux + Kp into band guidance: HF conditions (10/15/20m open/marg/shut), geomagnetic effect, auroral-VHF likelihood (6m/2m, beam N), D-layer absorption. Rule-of-thumb - **r** refresh - **⌵** back

MUF TO REGIONS (Space WX → m)

MINUMUF-3.5 path MUF from your QTH to 24 DX World regions: bearing, distance, MUF, best band, color-coded. F-region model, best 800-8000km - **⌵** scroll - **k** world map (colored dots) - **x** print - **⌵** back

WEATHER (menu)

Current conditions + multi-day forecast for your site (Open-Meteo). Refreshes on entry (WiFi) & with Update. Units in Settings - **r** refresh - cached offline - **⌵** back

QRZ LOOKUP (menu)

Callsign lookup via QRZ.com XML (needs QRZ XML subscription + user/pass in Settings → Network). **ENTER** type call → name/address/grid/class. WiFi req'd - **⌵** back

BAND PLAN (Help → f)

Worldwide amateur band definition LF→light; HF with ITU R1/R2/R3 splits, VHF/UHF/microwave EME+calling freqs, satellite subbands, IARU designators (H/A/V/U/L/S/C/X/K), sat modes incl QO-100. **⌵** scroll - **⌵** back

ACTIVATIONS (menu)

Upcoming sat activations on hams, at (roves, grid/special ops). List: date, call, sat, grid (c = your own entry). **⌵** move - **ENTER** detail **h** add your own sked (offline OK) **a** entry - **r** refresh - **⌵** back. Detail: UTC/mode/freq + **footprint** **note** (checks co-visibility with the activator +/-30 min of listed time) **⌵** scroll the full comment, a **SKED** reminder (T-60/30/10 beeps+flash), **w** multi-window screen if a footprint exists. Manual window: small polar plot (me + DX arcs), Date/AOS/LOS/dur + peak el each; **d** DX Doppler pre-set to the transponder & fixed DLUL parsed from the freq/comment (default table if none). Cached to card — last list shows offline; WiFi to refresh

OVERHEAD NOW (menu)

Snapshot of every catalog sat above the horizon right now, sorted by elevation, with az + rise compass dir (high=green, low=yellow) + count up/scanmed. **r** rescan (this instant) - **⌵** scroll - **⌵** back

TRANSPONDER DB (Sats → t)

Sat's transponder/beacon entries, two lines each: **D**=downlink+mode line, **U**=uplink+tone/rv line. Ordered two-way > amateur > active; inactive dimmed & **(off)**. **⌵** select (c = manual) - **x** del manual (2x) - **⌵** back

ORBITAL ANALYSIS

⌵ 11 pages: Info / Live / Next pass / Ground track / Doppler / Nodal / Sun-Beta / Pass outlook / Orbit position / Phys (velocity + launch date/age) / Explore (what-if apo/per/incl sandbox, x reseed) / r recompute - z orbital-zone transits - Doppler f sets beacon freq

ORBITAL ZONES (Orbital → z)

When the sat transits distinctive regions, w/ enter/exit times + duration + live in/out + L-shell. z cycle zone: **SAA** / **Eclipse** / **Polar** (>60°) / **Inner** & **Outer Van Allen belts** (L-shell, higher orbits) - **⌵** scroll - **x** print - **⌵** back

SIMULATION

⌵ step time - **⌵** step size - **m** world-map view (sub-point + footprint) - **x** reset to now - **⌵** back

LOCATION

e/o/a lat/lon/alt - **g** grid - **p** GPS on/off - **s** GPS source - **c** set clock - **v** live position (DMS) - **ENTER** GPS spy plot

GPS SKY PLOT

Live GNSS by az/el, colored by signal (green=strong, gray=weak) - **⌵** back

WORLD MAP

All footprints + night-side shading - **f** highlight favorite - **c** recenter on QTH/0° - yellow=sunlit cyan=eclipse - **⌵** back

SCHEDULES

10-day: **⌵** scroll +/−1 day (fills full days), **r** recompute - illum: **⌵** scroll +/−60 days - Mutual: **⌵** scroll, **ENTER** polar detail, **d** Doppler

LOG

Menu (scrolls): **ENTER** new QSO / browse / export ADIF / LoTW upload / Cloudlog upload / voice memos / notes / awards / fill grids (QRZ) - **Awards**: QSO/grid/date/DXCC totals (states+DXCC deleted from grid), **g/s/d** scrollable worked lists, **ENTER** per-sat - List: **⌵** scroll, **ENTER** edit - Entry: **⌵** live, **ENTER** edit, **s** save, **x**2 delete - editing a QSO re-arms its upload; extra **LoTW/Cloudlog** signs (ENTER toggles) override that

LOG & UPLOAD TO LoTW (Log → Sign & upload)

Uploads sat QSOs direct to ARRL LoTW over WiFi. Needs SD + your LoTW key (lotw_key.pem/lotw_cert.pem in /CardSat/ → make them from your QSL p12 with tools/lotw_cert_converter.html in a browser; in Settings pick DXCC → primary (state/province/...) →

secondary (county/city) + zones + IOTA, all gated pickers w/ type-to-filter) — see manual §8. **u** upload - **a** re-send all - **⌵** back

UPLOAD TO CLOUDLOG (Log → Upload to Cloudlog)

Uploads sat QSOs to a self-hosted Cloudlog/Wavelog over WiFi (an alternative to LoTW — Cloudlog can forward to LoTW). Set URL + read-write **API** key + **station ID** in Settings. **u** upload - **a** re-send all - **⌵** back

VOICE MEMOS (Log → Voice Memos)

Browse newest-first (date/time/sat/len). **ENTER** play - **n** new - **d** del - **r** refresh - record via **v** on Track. SD req'd

NOTES (Log → notes)

Free-form text notes (txt in /CardSat/notes/, newest-first w/ date/time). Browser: **ENTER** open - **n** new - **d**ENTER del. Editor (commands use **F** no **⌵** type): type freely, **ENTER**=newline, **DEL**=backspace **F**nt/**F**nv cursor L/R **F**nt/**F**nv up/down **F**nv+ save - **exit** (auto-saves)

LORA MESSAGES (Home → Messages)

CardSat-to-CardSat broadcast chat (Cap LoRa). Same freq/SF/BW = same group. Set **region** (US 33cm / EU 70cm / JP 430) in Settings for a legal default freq, n write/send - **⌵** scroll+select (newest on bottom) - **o** station roster - **r** retry - **m** LoRa RX/hex monitor - **⌵** back. **Actionable mssgs** (plain text, interop): a msg with @lat,lon + **SAT** / **ISAT** date time → **ENTER** opens bearing/hex monitor (dist/bgr/grid) / sat detail / pre-filled sked; **rsAT** carries NORAD (#name/norad) so it resolves across differing names. Send for the current sat & your location: **p** position (also a presence ping) - **s** satellite - **k** sked (date→time). **Roster** (o): who is heard, with grid + dist/bgr + signal; **ENTER**=compass, **p**=ping. Opt-in **Auto position** reply setting answers others' positions (look-guarded). **Rx always** on: badge + banner (opt-in beep, **Msg notify** in Settings). **Share GP elements**: a peer's **SATS** → **L** pops an import **satellite?** prompt (sender/name/NORAD) — **y** add/update, **n** decline; checksum-verified, experimental. Needs RadioLib build.

LORA RX / HEX MONITOR (Messages → m)

Receive/inspect any LoRa signal (not just sat's). Config: set freq (type in **MHz**), SF, BW, CR, sync, preamble, CRC — saved across reboots. **ENTER** starts RX. Monitor: live hexdump+ASCII, RSSI/SNR, **p** pause (read a frame on a busy channel) - **⌵** scroll - **s/b/c/f** tune SF/BW/CR/step - **⌵** freq - **⌵** esc. Rx-only. Untested HW

UPDATE

kENTER GP+clock+AMSAT+space-wx+weather - **f** fast (GP+AMSAT+favs' TX) - **a** cache all TX (auto-reboots) - **w** WiFi only

SETTINGS

⌵ change - **ENTER** edit/toggle - **s** scan WiFi - opt. 2nd WiFi (field fallback) - reset = ERASE

DUAL RIG (2 radios)

CAT type dual = a downlink + an uplink radio driven natively (27 radios). **Settings → Dual-Rig** setup: per leg pick **CAT** rig; **LAN** bus (Grove / USB / LAN), **CI-V**, baud (**Grove honors CI-V wiring**, incl. 1-pin); a leg set to **None** = that half simply not CAT driven; **LAN** legs take the radio's IP - port (**CI-705 over its own WiFi**; **u** ping); a nominate USB adapter (both legs may be USB via a hub) - **s** save+audio. One Grove leg only; PTT stays manual; FM tones set on the radio. USB helper (Grove): a 2nd USB host on an M5stickS3 for a radio or rotorator.

GP SOURCE

AMSAT / **CelesTrak** JSON-PP category / **Custom URL** - **⌵** move - **ENTER** select

ROTATOR (manual)

⌵ az - **⌵** el - **s** step - **x** stop - **⌵** back

NETWORK SERVERS

rigctld PC drives rig (VFOA/DLB=UL) - **rotctld** PC drives the configured rotator - **rigctl** drives remote rig - **Web** opt-in mobile page (no auth, trusted LAN); live sky plot. Doppler readout, tap-to-copy freqs, visible-pass list + AOS alerts, radiolocator control, **Files** = download-only /CardSat browser (no upload)

EDIT

type - **DEL** backspace - **ENTER** ok - **⌵** cancel

NEARBY & DX (Home)

Live-feeds hub. **APRS heard**: live APRS-IS listen (needs calling; **g** center grid, **ENTER** bearing detail, **f** restart). **DX cluster**: **f** fetch, **n** next band with spots, comments shown. **ADS-B radar**: polar range plot, **f** fetch, **t** scatter target grid; source URL in Settings > Network. **QRZ lookups**, **p** prints each feed.

ABOUT

Build/version, IP, free heap and diagnostics (read-only). **r** Station readiness checklist - **t** **Tools** (63): scientific/graphing/programmer calculators, **Tiny BASIC** (A-Z numbers, AS-Zs text, LEFTS/MDS/INSTR, DIM arrays, **INPUT** form before the run, SATSELFPRINT/graphics), **location converter** (grid/DMS/DDM/Plus/UTMMGRS), DXCC/CQITU lookups, RF/antenna workshop, band budget, phasing/sub, attenuator, RF exposure, orbit lifetime, **State vector** → GP, Q-codes/phonetics/RST, CTCSS, **Telnet** (LAN terminal, 10 saved hosts, opt. printing), **satbuild** tools: conjunction screener, orbital neighborhood, transponder planner, link-margin curve, debris-group screen, Doppler badge, cascade NF/GIT, sun-noise G/F, helix, L/P/U/T match, pointing loss, **IM0**, microstrip 20, toroid, delta-v, thermal, Faraday, ampacity, PLL plan - **p** **Print**: 30 reports to network printer / serial / 80-cel / Reports file (any mix); contextual **m** on report screens + all form tools, **p** all-passes & polar map - **i** License & credits - **z** **Games menu**: seven mini-games (**⌵** pick, **ENTER** launch).